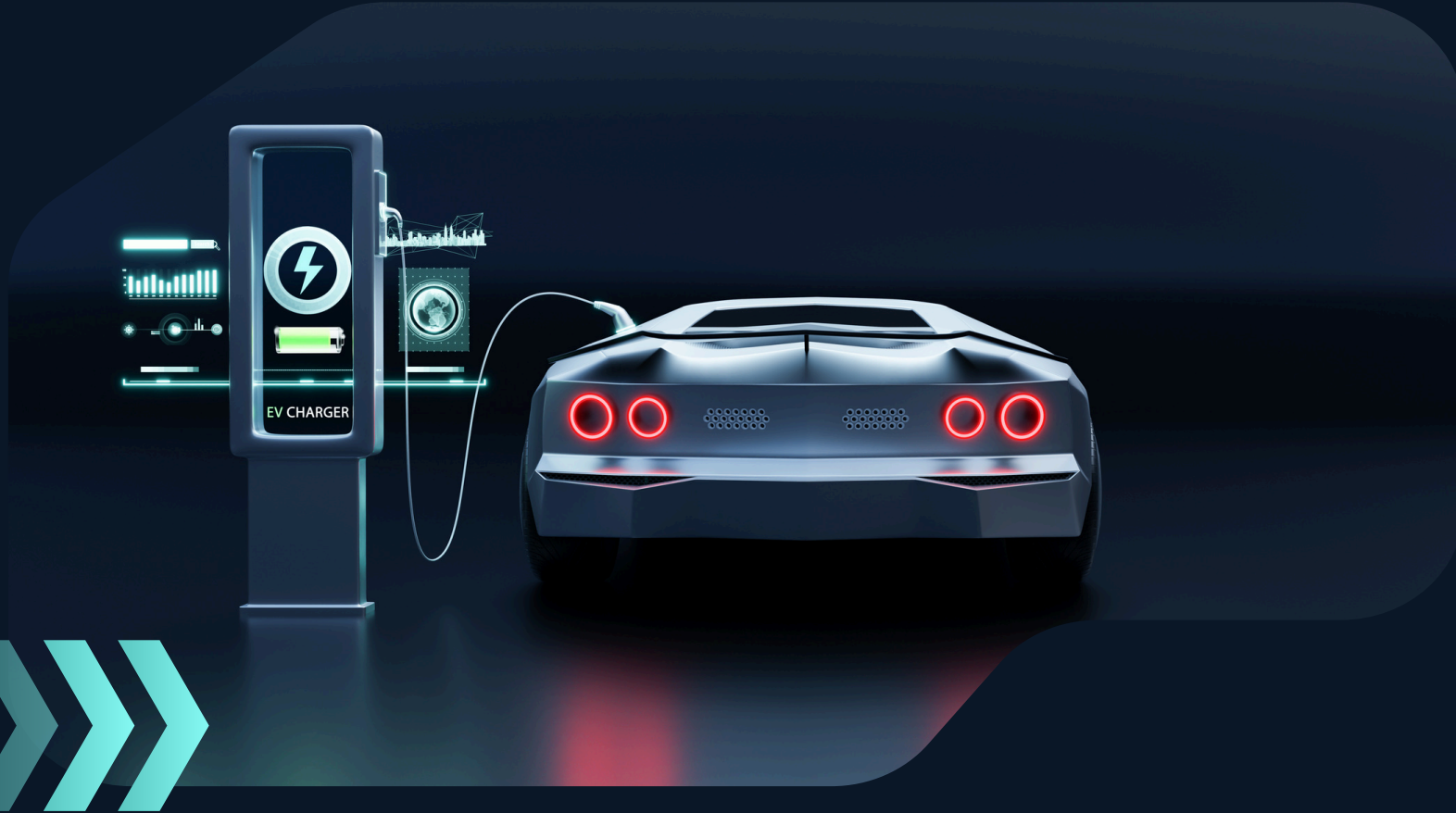
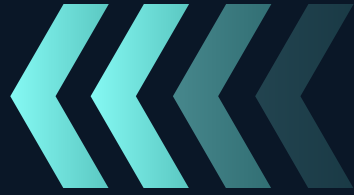


BMW GLOBAL SALE



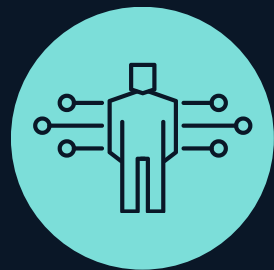
CHANOUDOM TANN 6610755487

APIVIT DENPRUKTHAM 6610755354



WHY THIS DATA

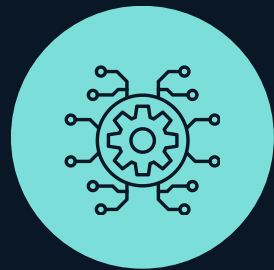
BMW Global Sale Dataset



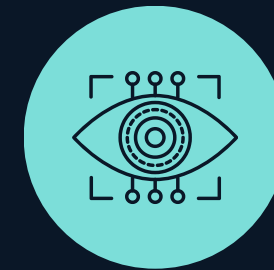
Structured and clean data



Practical real-world use



Suitable for ML modeling



Interesting classification challenge



INTRODUCTION

To The Dataset

This data consists 11 columns and 50K entries.

- Model: Name of the BMW vehicle Model
- Year: Manufacturing or Sales Year
- Region: Geographic market where the vehicle was sold
- Color: Vehicle Color
- Fuel_Type: Type of fuel used (Petrol, Diesel, Hybrid, etc.)
- Transmission: Transmission type (Manual or Automatic)

BMW sales data (2010-2024)

Model	Year	Region	Color	Fuel_Type	Transmission	Engine_Size_L	Mileage_KM	Price_USD	Sales_Volume	Sales_Classification
5 Series	2016	Asia	Red	Petrol	Manual	3.5	151748	98740	8300	High
i8	2013	North America	Red	Hybrid	Automatic	1.6	121671	79219	3428	Low
5 Series	2022	North America	Blue	Petrol	Automatic	4.5	10991	113265	6994	Low
X3	2024	Middle East	Blue	Petrol	Automatic	1.7	27255	60971	4047	Low
7 Series	2020	South America	Black	Diesel	Manual	2.1	122131	49898	3080	Low
5 Series	2017	Middle East	Silver	Diesel	Manual	1.9	171362	42926	1232	Low
i8	2022	Europe	White	Diesel	Manual	1.8	196741	55064	7949	High
M5	2014	Asia	Black	Diesel	Automatic	1.6	121156	102778	632	Low
X3	2016	South America	White	Diesel	Automatic	1.7	48073	116482	8944	High
i8	2019	Europe	White	Electric	Manual	3.0	35700	96257	4411	Low
3 Series	2012	Middle East	White	Electric	Manual	2.4	101595	117995	500	Low
i8	2016	North America	Blue	Hybrid	Manual	1.9	116863	74539	8252	High
7 Series	2020	North America	Silver	Diesel	Automatic	3.8	27403	100015	8111	High

- Engine_size_L: The volume of all cylinder in engine ,measured in Liters(L)
- Mileage_KM: Distance driven in kilometers
- Price_USD: Vehicle price in US dollars
- Sales_Volume: Number of units sold
- Sale_Classification: High and Low

RESEACH QUESTIONS

Can BMW models be accurately classified using the features available in the dataset?

Does the color of the car affect sales in each country?

What is the overall performance (accuracy, precision, recall) of the classification model?

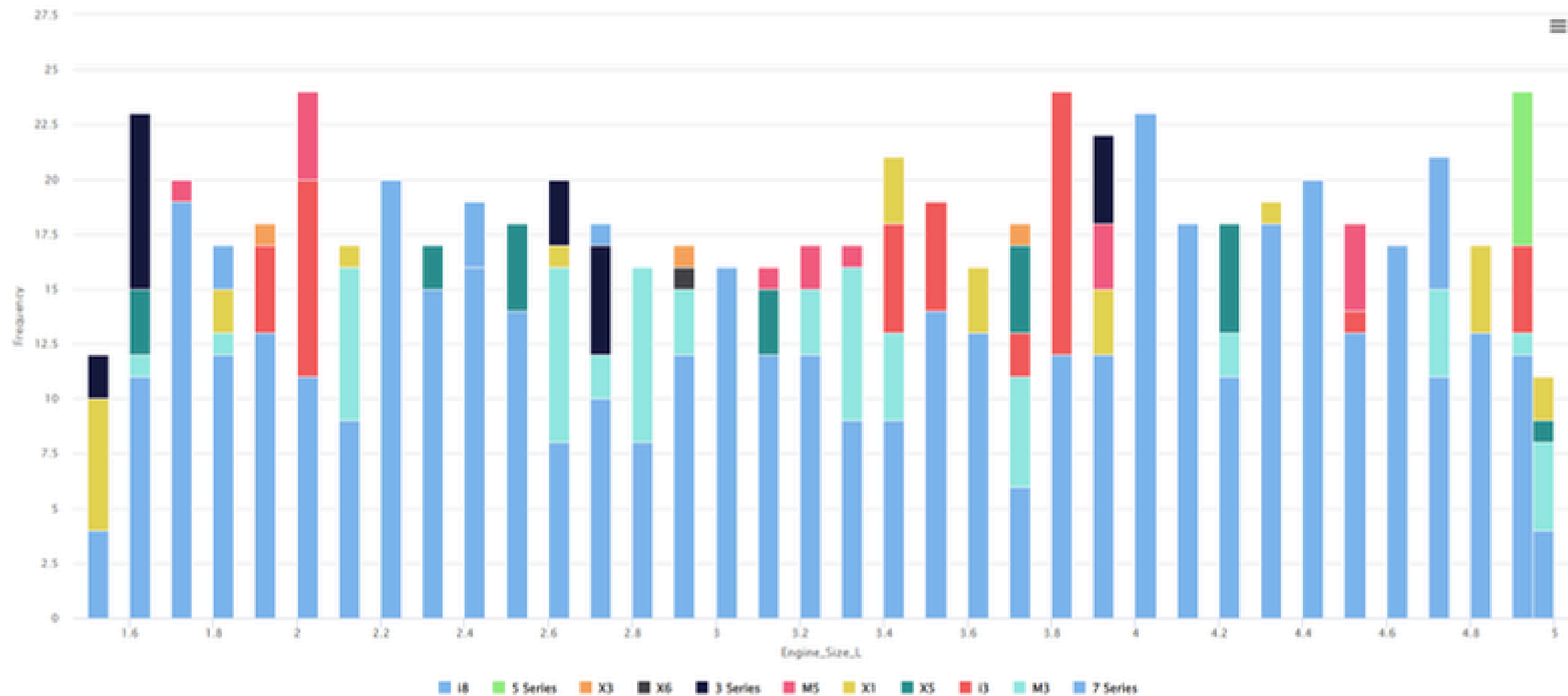
HYPOTHESIS

Certain features significantly distinguish BMW models.

(Engine_size, Transmission, Fuel_type, Model)

Car color significantly affects sales across different countries.

(Color, Volume_sales, Country, Model)



THANK YOU FOR
YOUR ATTENTION

